

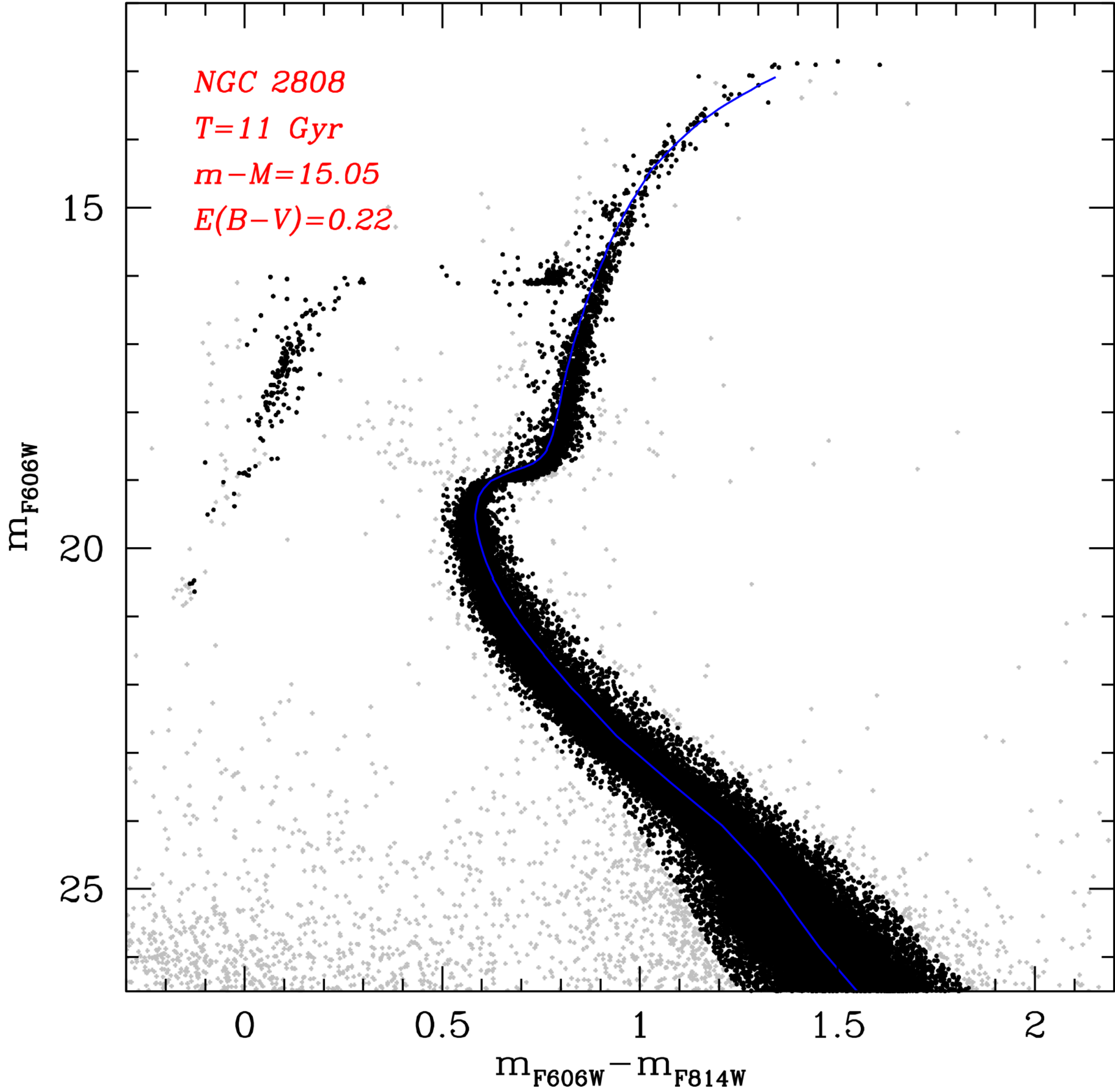
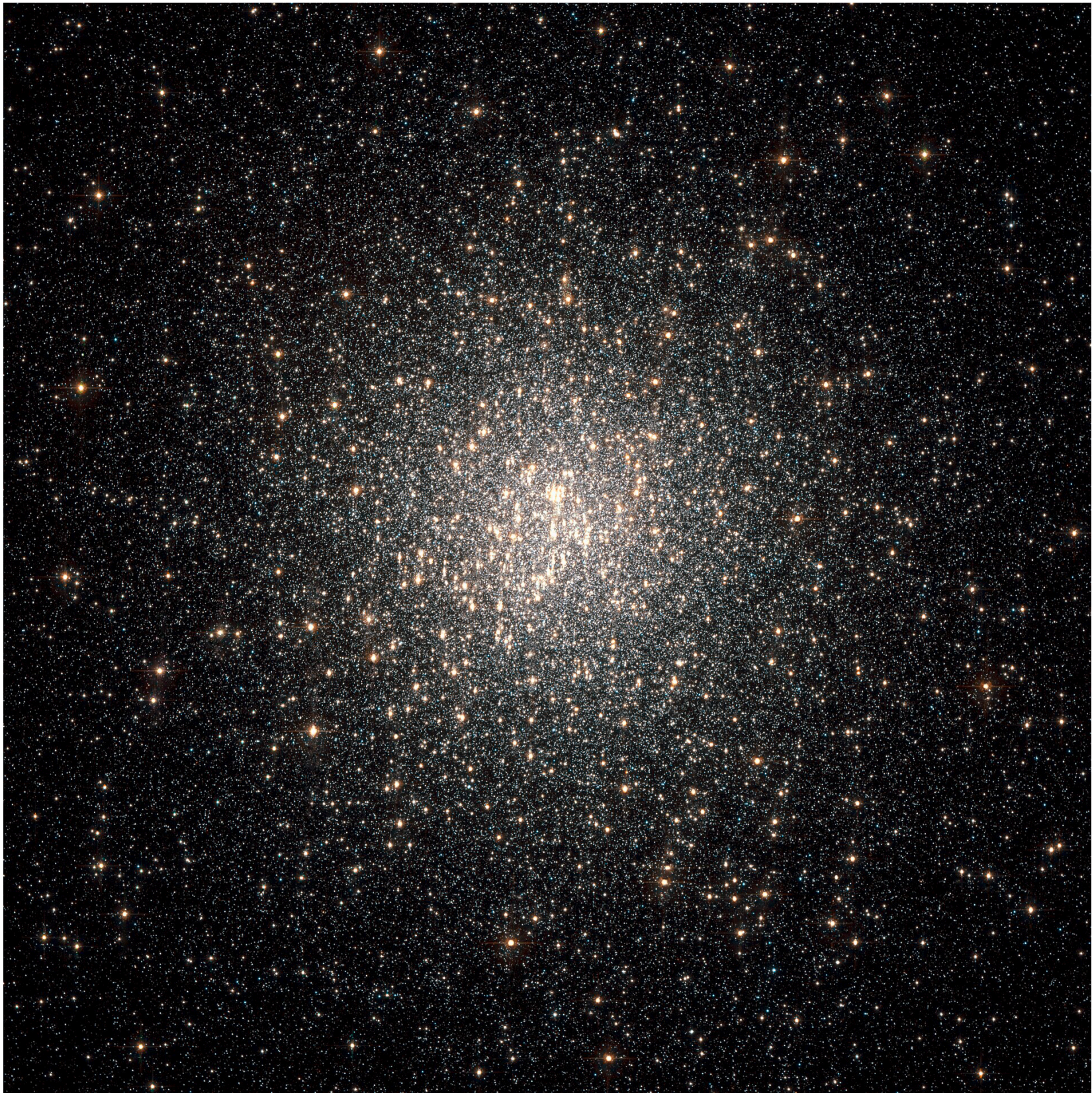
Multiple Populations in Globular Clusters

AY 250: Stellar Populations

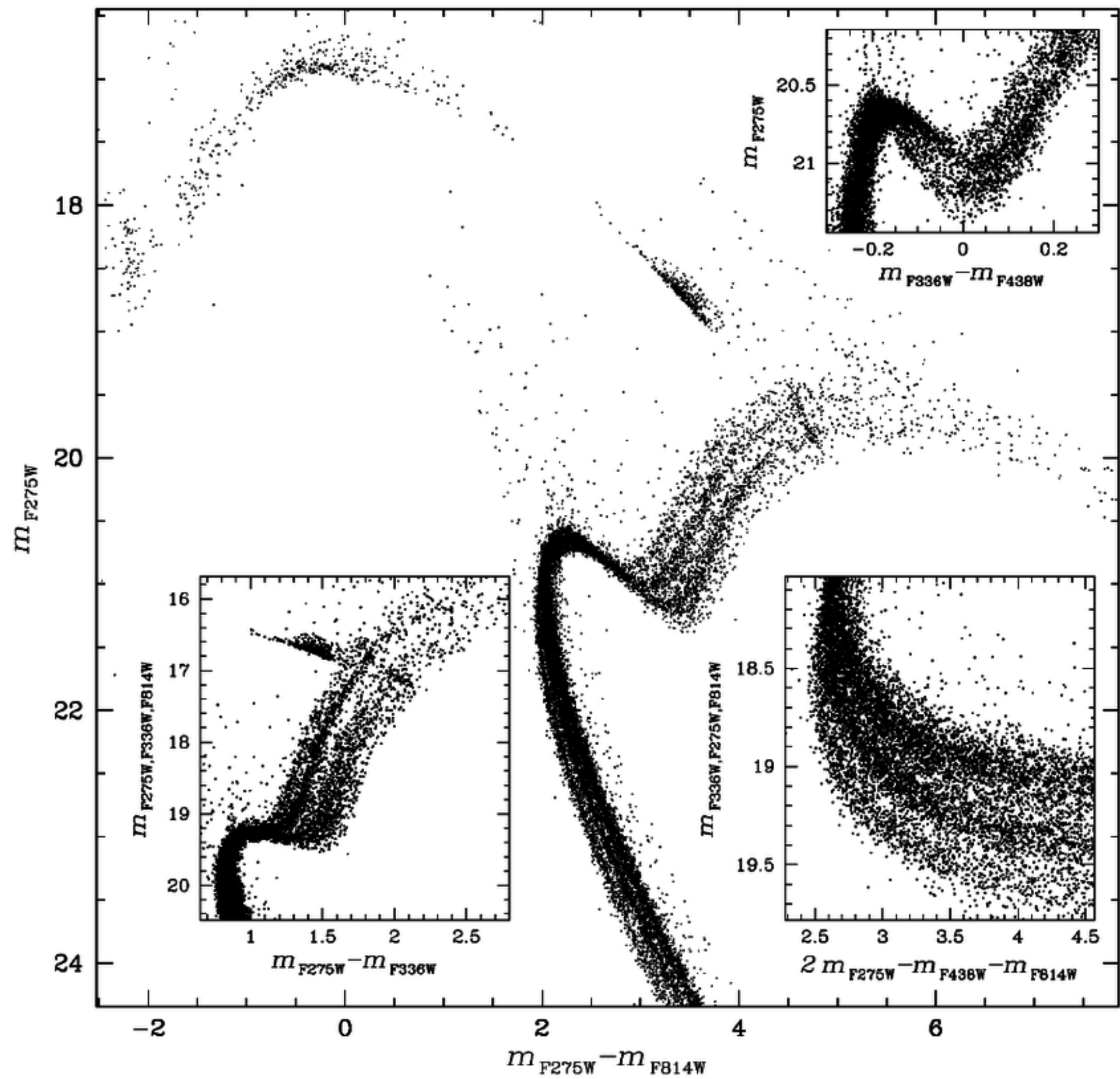
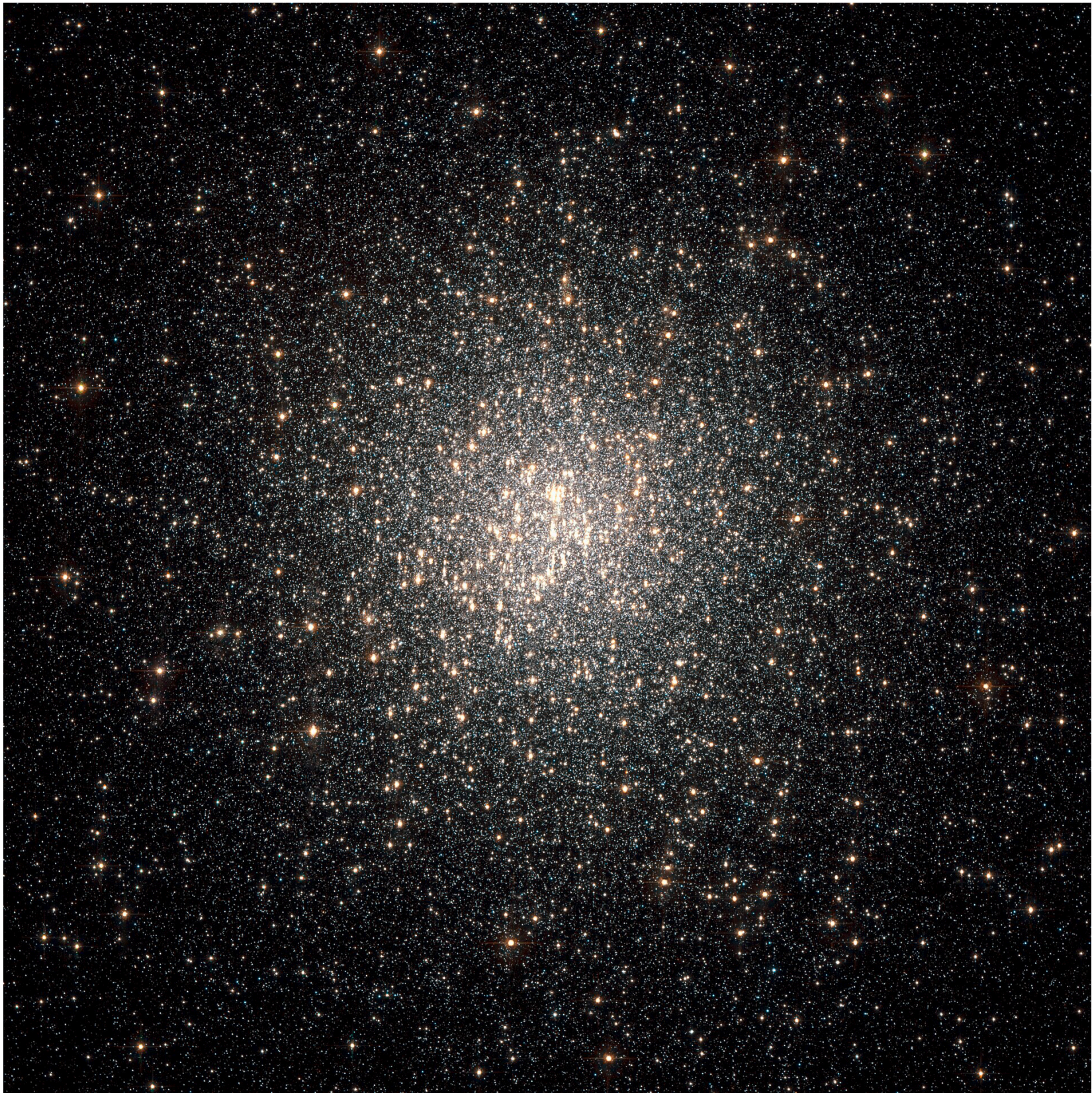
Feb 26, 2026

Dan Weisz

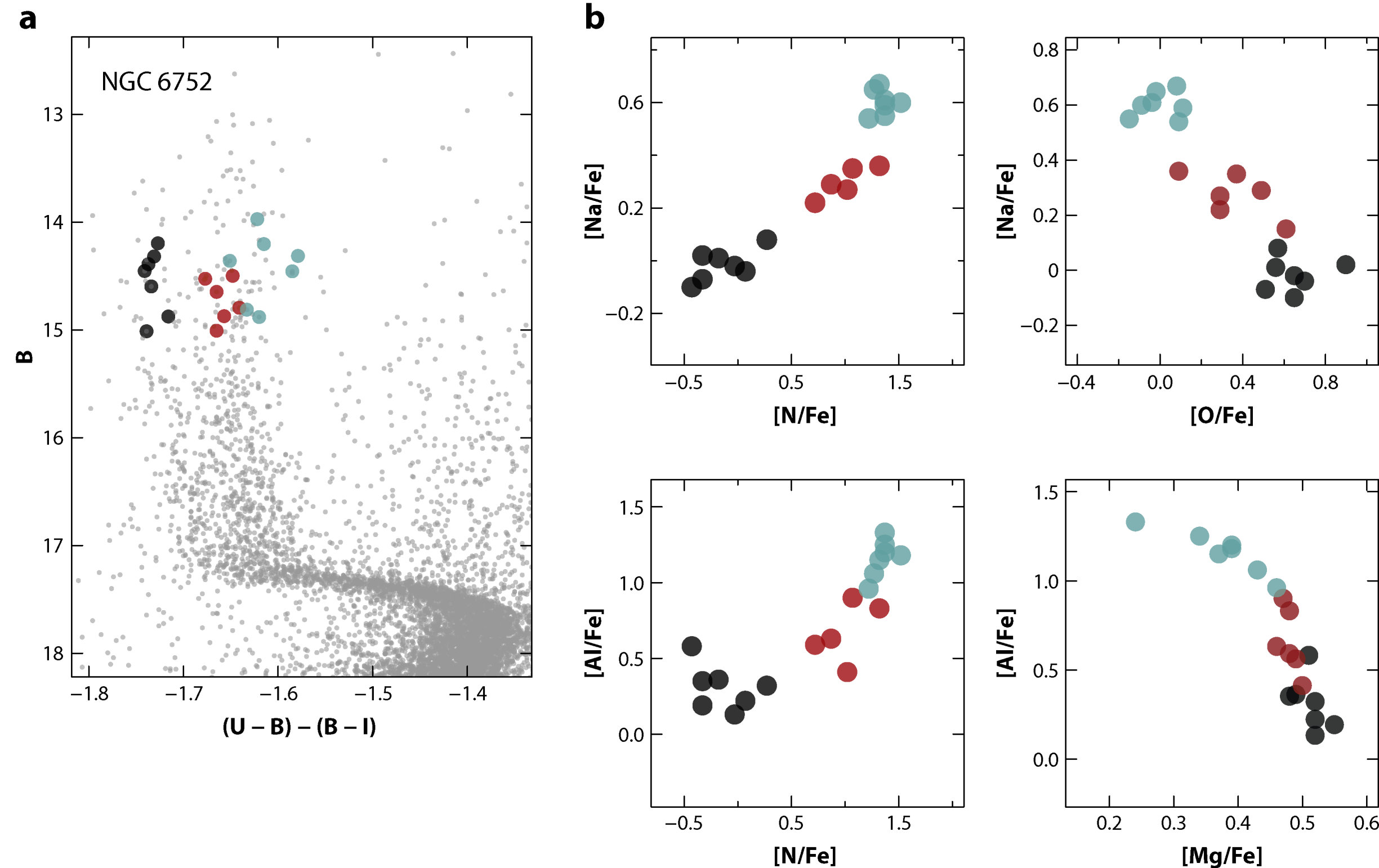
Optical Color-Magnitude Diagram



UV/optical Color-Magnitude Diagram



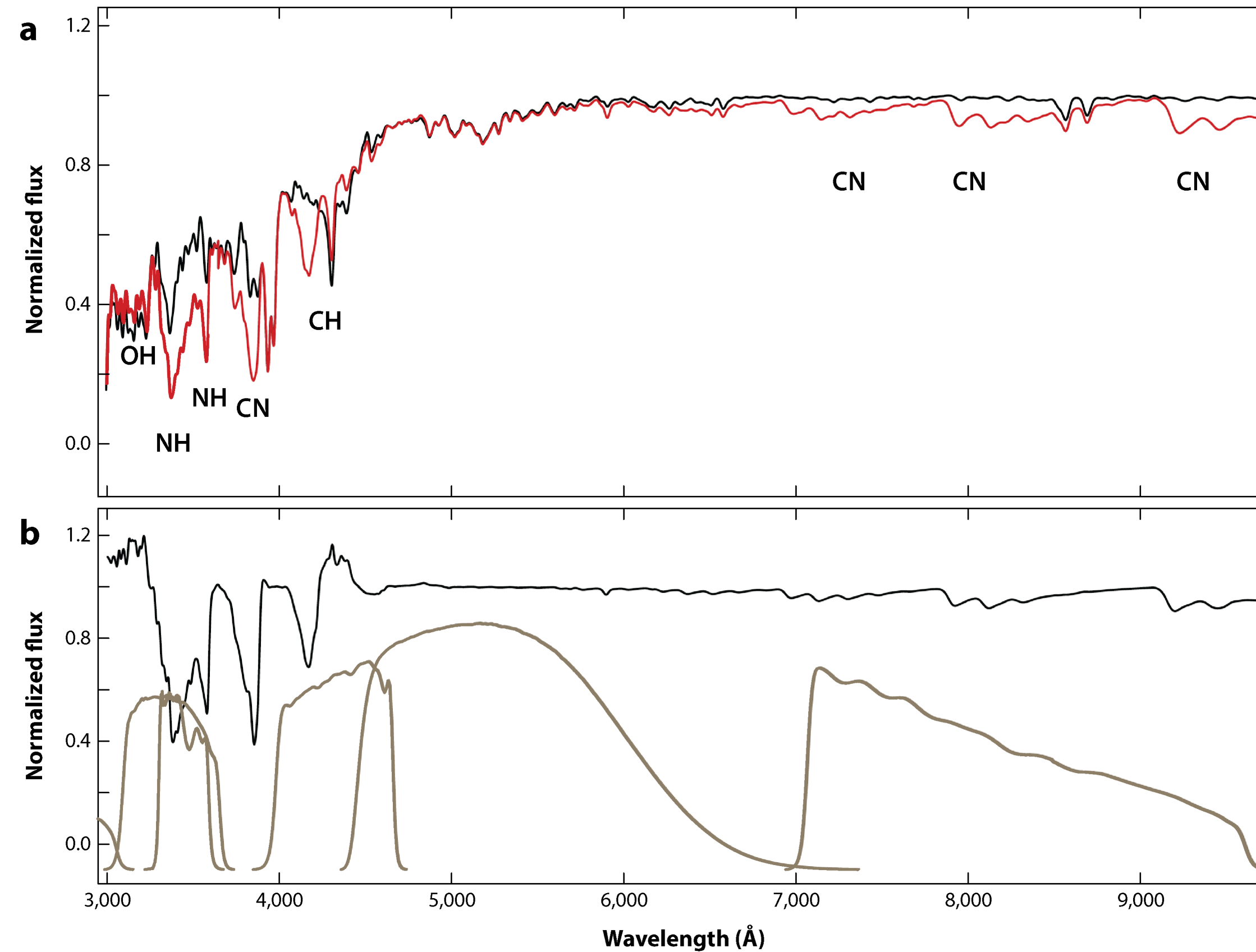
Chemically Distinct Populations in GCs from Spectroscopy



Sodium-Oxygen (Na-O) anti-correlation: within a single cluster, some stars are rich in oxygen and poor in sodium, while others are rich in sodium and poor in oxygen.

Similar anti-correlations were found for Magnesium-Aluminum (Mg-Al) and Carbon-Nitrogen (C-N). Crucially, heavier elements like Iron (Fe) remained mostly constant.

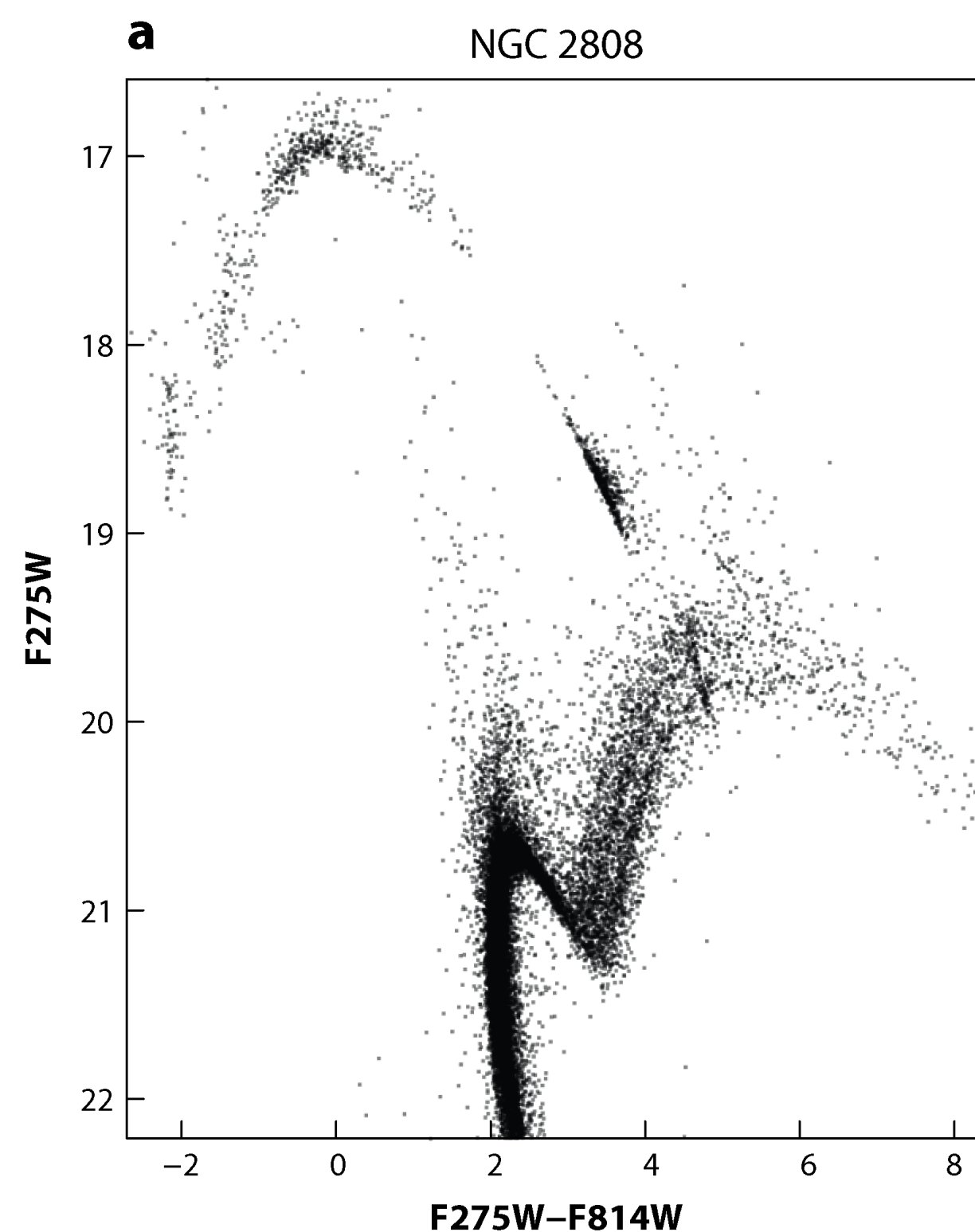
Why is photometry sensitive to chemically distinct populations?



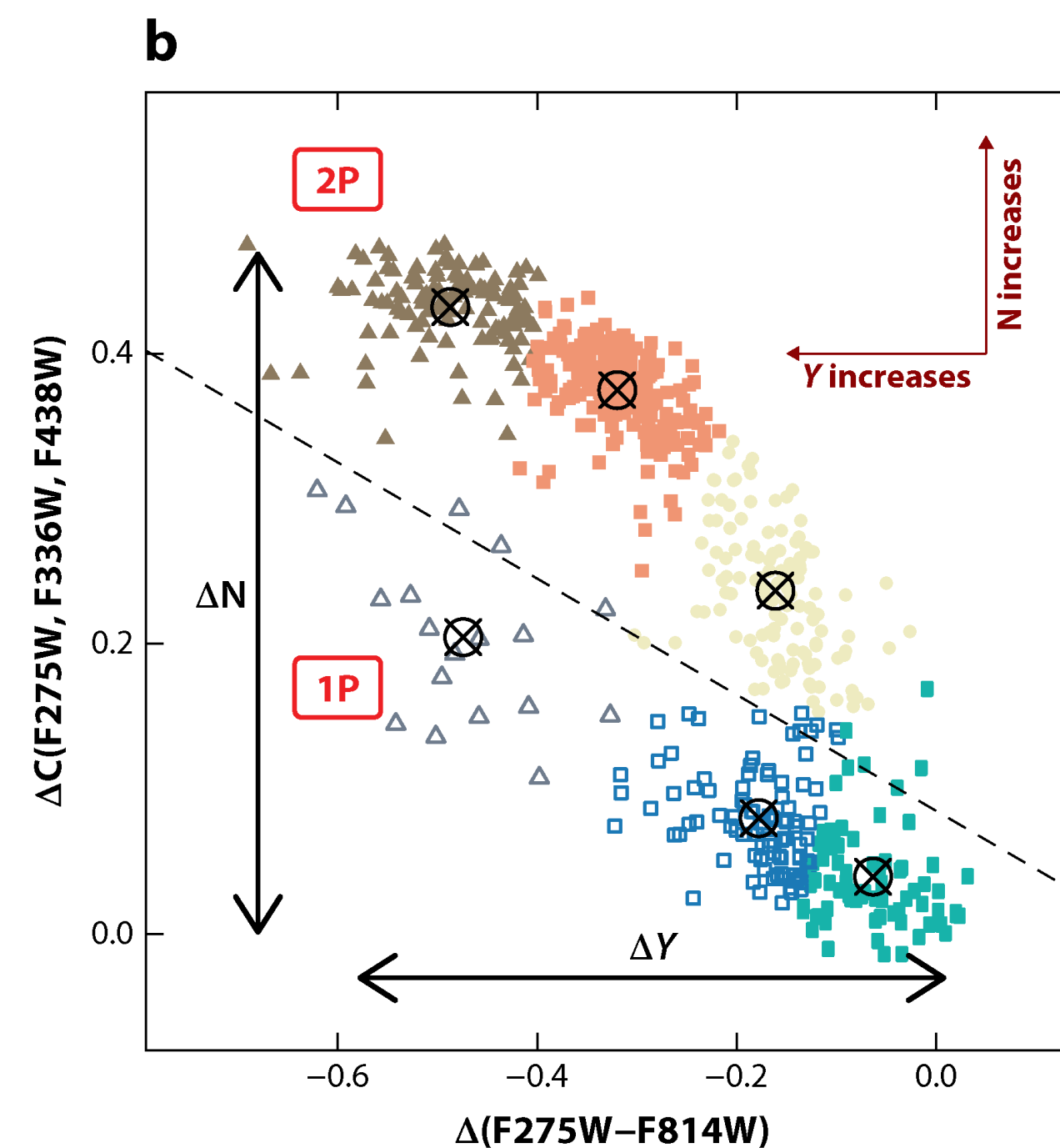
 Bastian N, Lardo C. 2018.
Annu. Rev. Astron. Astrophys. 56:83–136

Helium enrichment changes opacity, which leads to hotter temperatures for fixed stellar parameters (e.g., mass, surface gravity). Detections of He enhancements are clearest at UV wavelengths.

“Chromosome Maps”



Bastian N, Lardo C. 2018.
Annu. Rev. Astron. Astrophys. 56:83–136



1P Stars (The "Pristine" Group): Their chemistry matches the normal, background stars of the Milky Way's halo. They have **high Oxygen and Carbon**, and **low Sodium and Nitrogen**.

2P Stars (The "Polluted" Group): These stars were born from gas that went through high-temperature hydrogen fusion. They have **low Oxygen and Carbon**, but are **highly enriched in Sodium, Nitrogen, and often Helium**. In extreme 2P stars, Magnesium is depleted while Aluminum is enriched.

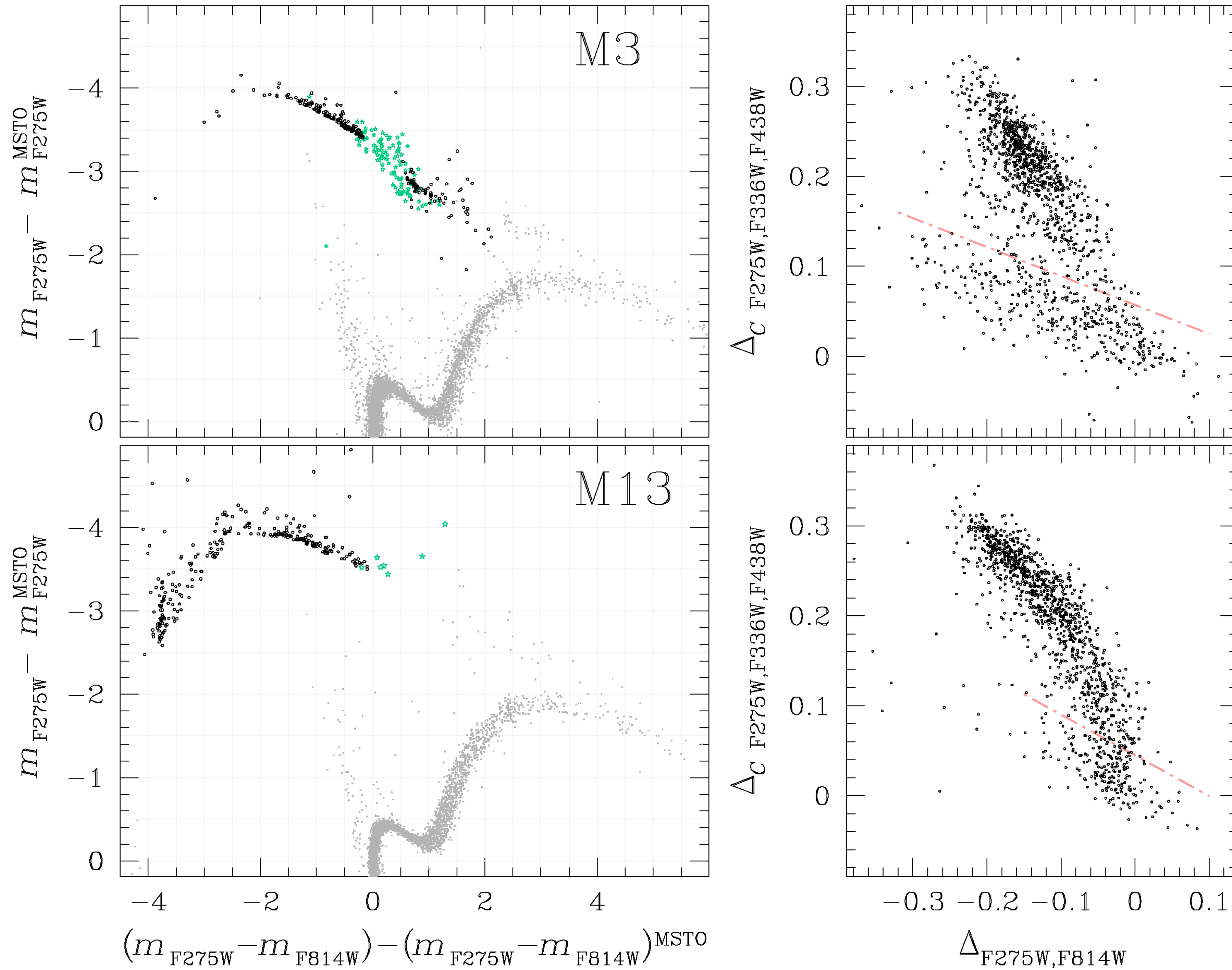
1P Stars: Typically make up only **20% to 50%** of the cluster's current mass.

2P Stars: Are the dominant group, making up **50% to 80%** of the stars in the cluster.

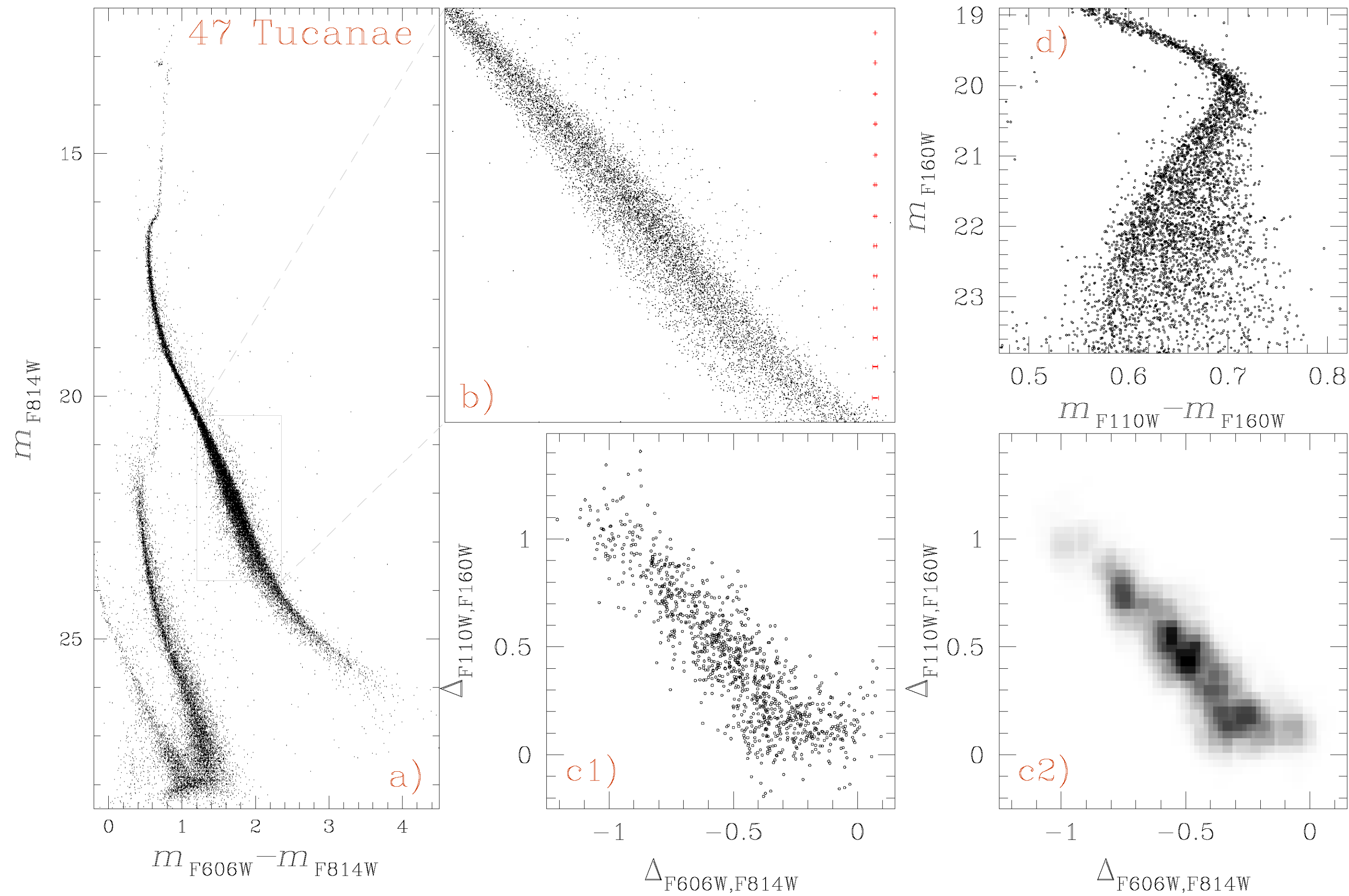
1P Stars: Tend to be more dispersed and populate the **outer outskirts** (the halo) of the globular cluster.

2P Stars: Are highly **centrally concentrated**. They are packed tightly into the dense core of the cluster.

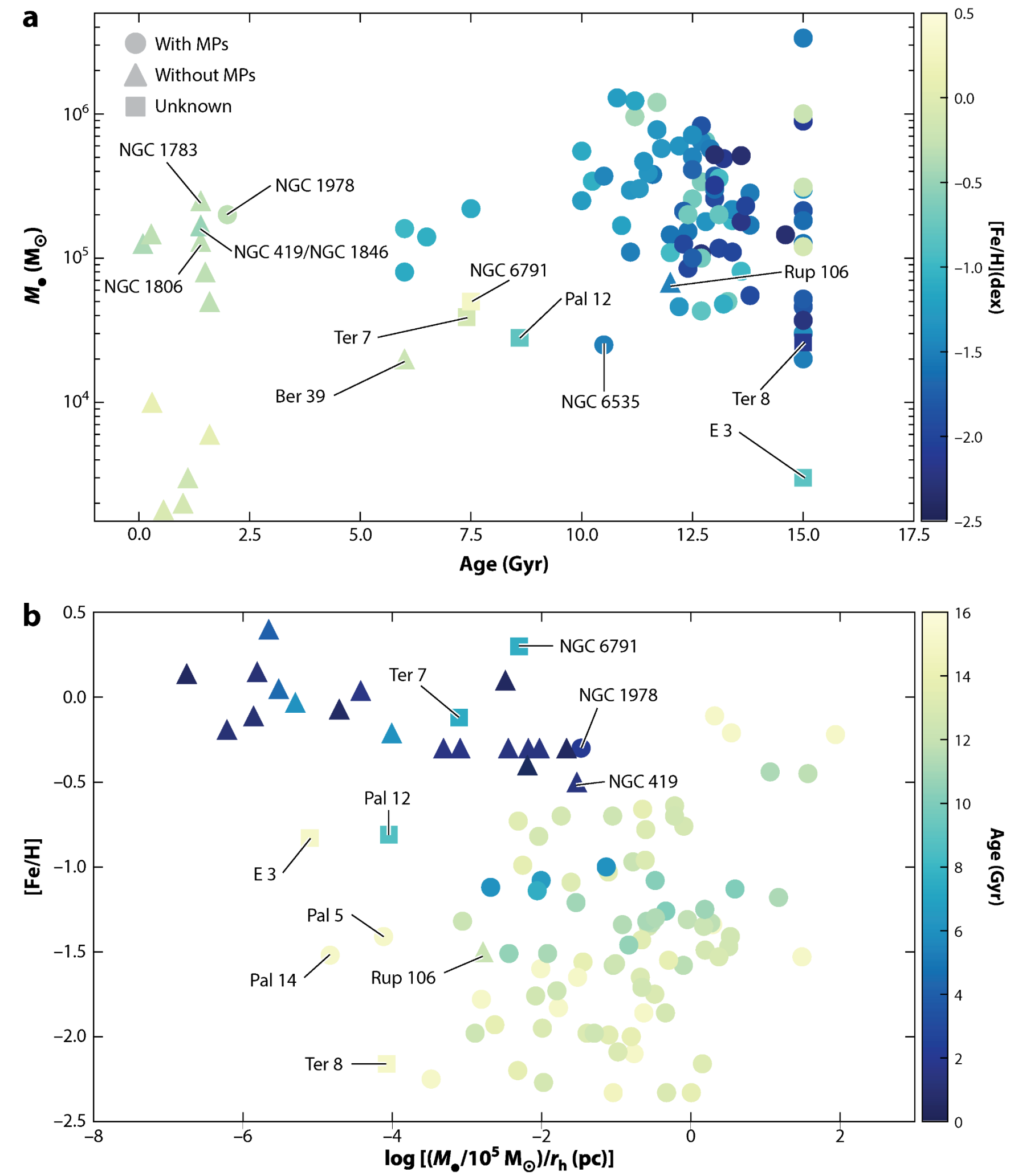
“Chromosome Maps” are different among GCs



Multiple Populations also be detected in the IR for cools stars



Multiple Populations Trends in Milky Way Globular Clusters

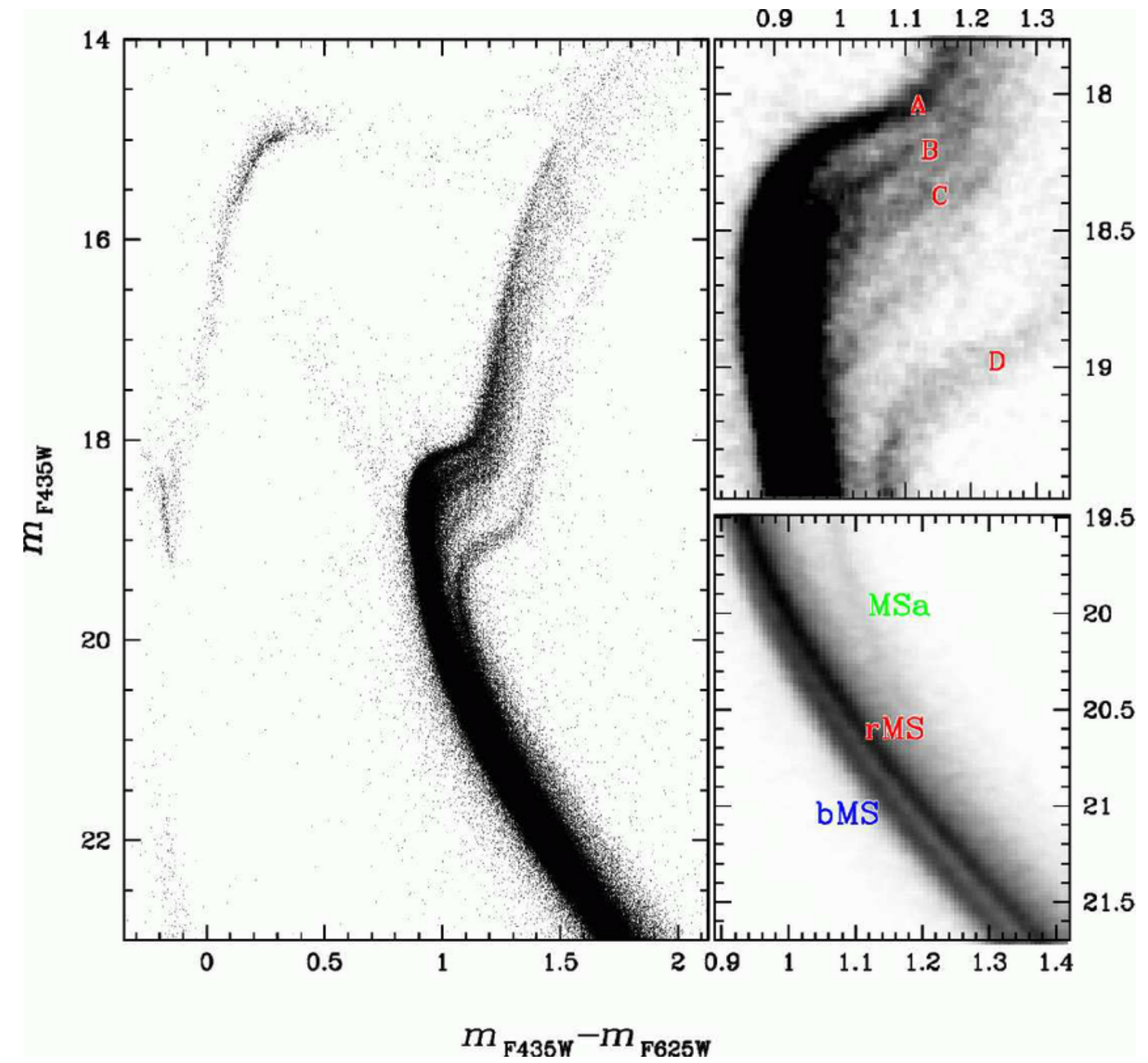
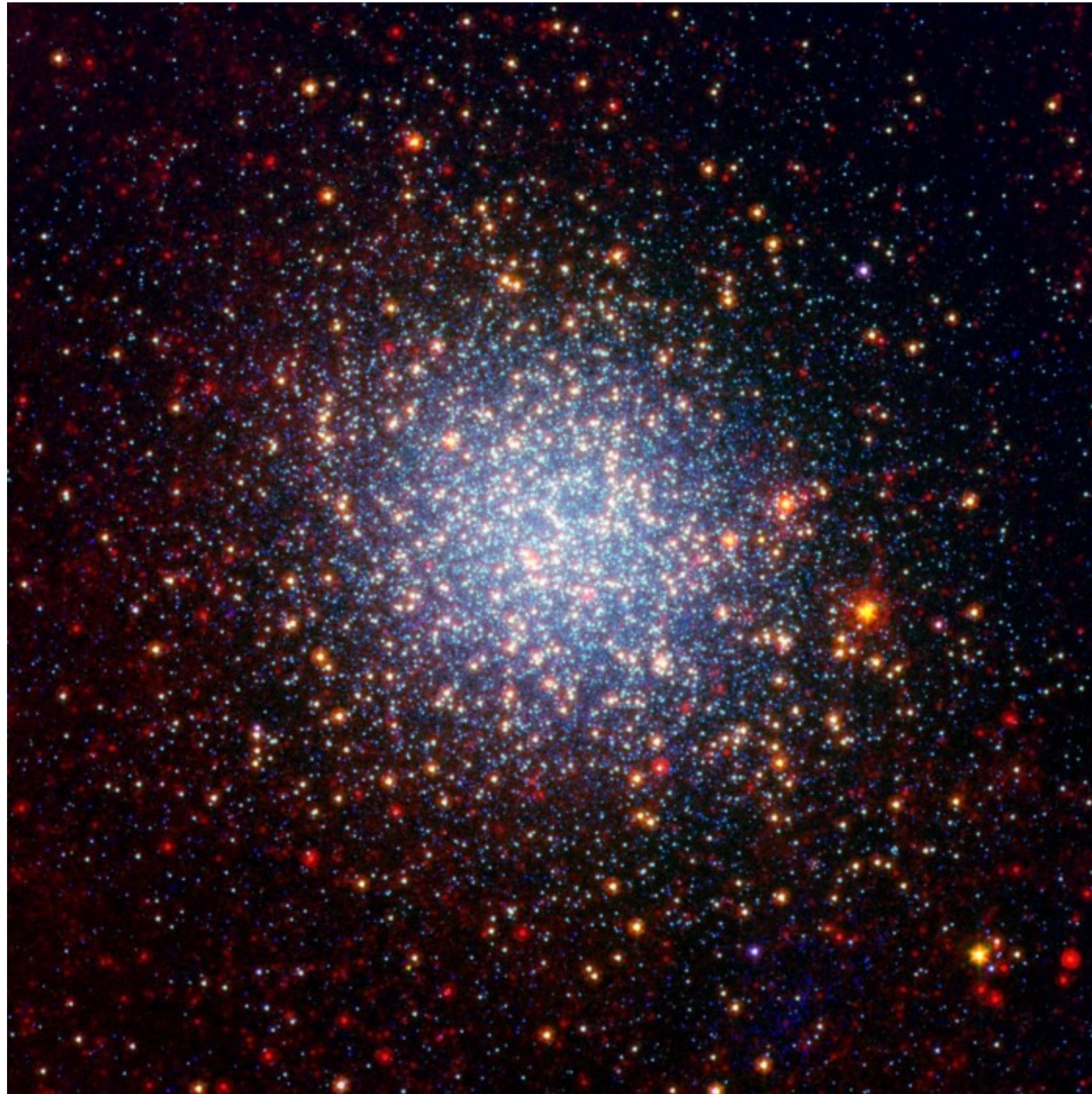


Multiple Populations Summary

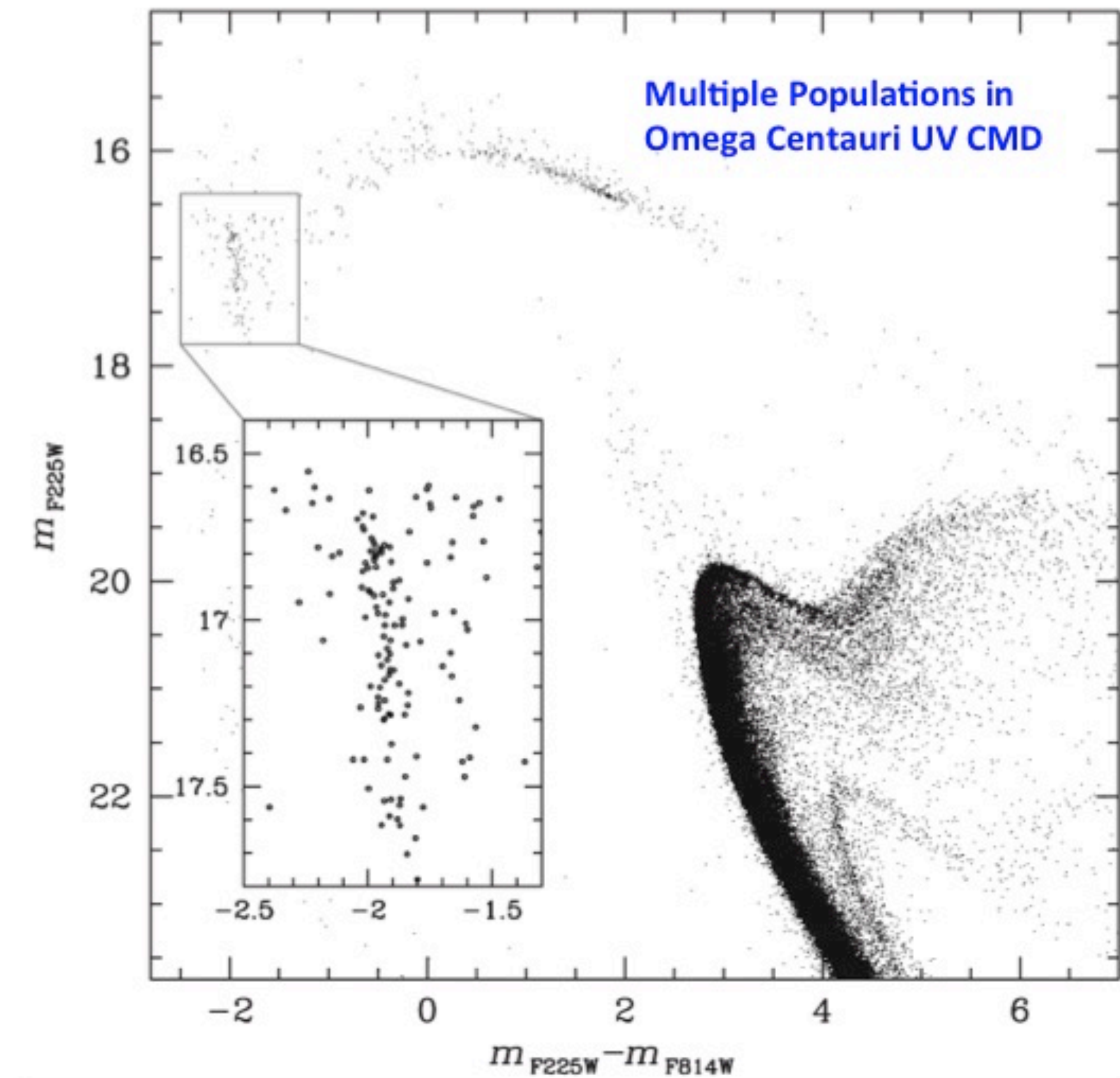
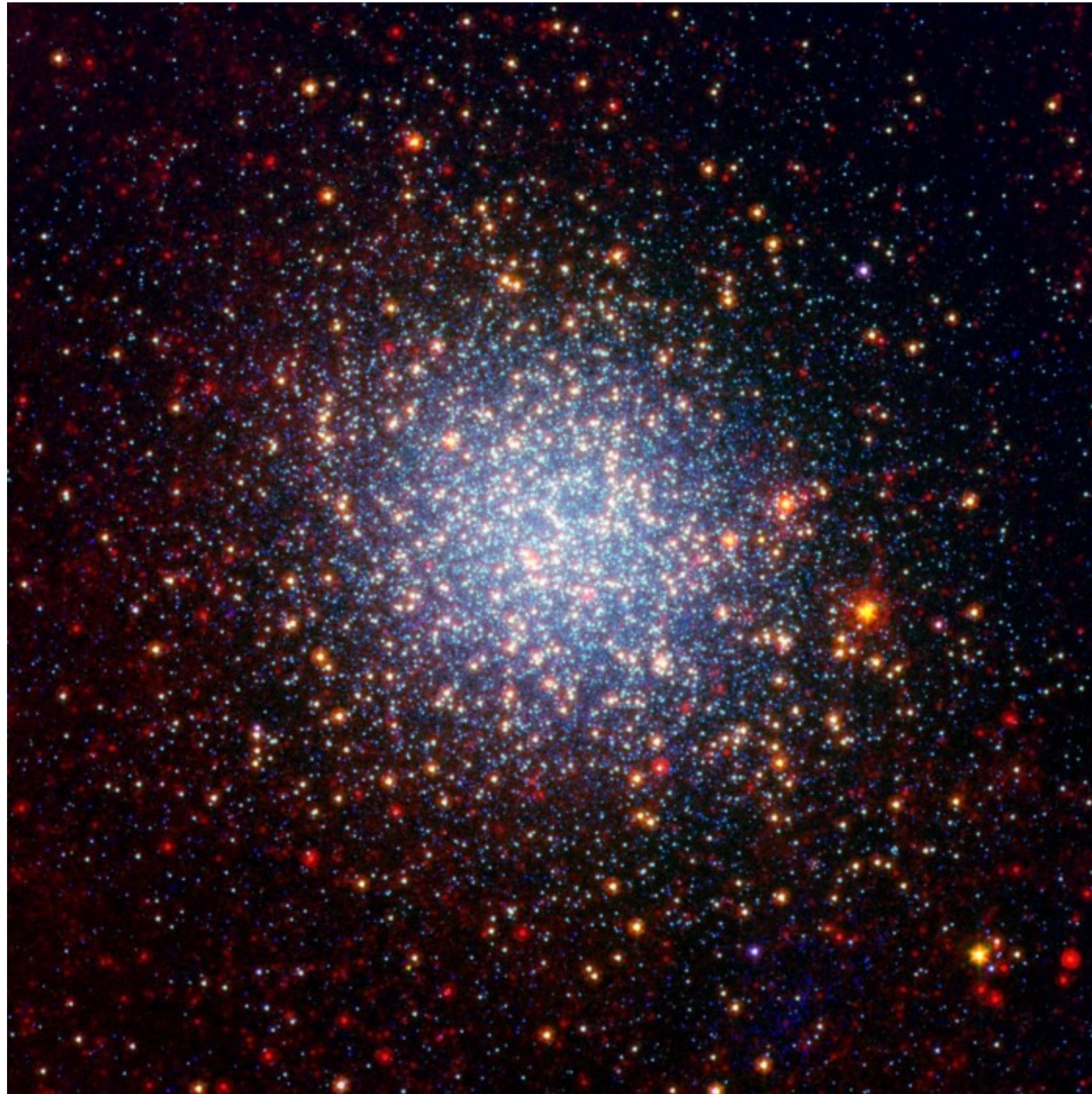
1P vs. 2P Stars in Globular Clusters

Feature	1P (Primordial)	2P (Enriched)
Chemistry: Oxygen & Carbon	High (Halo-like)	Low (Depleted)
Chemistry: Sodium & Nitrogen	Low (Normal)	High (Polluted)
Helium Content	Normal (~24%)	Enhanced (Up to ~38%)
Spatial Location	Cluster Outskirts	Central Core
Total Population	Minority (20% – 50%)	Majority (50% – 80%)

The Curious Case of Omega Cen



The Curious Case of Omega Cen



Theoretical Scenarios for why Globular Clusters Have Multiple Populations

	(Near) ubiquity (Section 2)	Abundances (Section 5.1)	Li correlations (Section 5.1.2)	Variety/Stochasticity (Section 5.1.3)	Mg–Al (Section 5.1.5)	Discreteness (Section 5.2)	Mass budget (Section 5.2)	f_{enriched} correlation with GC mass (Section 5.4)	He spread correlation with GC mass (Section 5.5)	YMCs (Section 5.5)	Age trend (Section 5.6.1)
AGB	X	X*	X*	X	✓*	X*	X*	X	X*	X	X
FRMS	X	X*	X	X	X	X	X*	X	X	X	X
VMS	X	X*	X	?	X*	X	✓*	✓*	✓*	✓	X
EDA	X	X*	X	X	X	X	X*	X	X	✓*	X
Reverse order	X	X*	X	?	?	X	✓*	X	X	X	X
eSF period	X	X*	X	X	X	X	X*	X	X	✓	X